

API Design Example

 systemdesignschool.io/fundamentals/api-design-example



Let's put what we've learned from the previous section into practice and design a REST API for a Twitter(X)'s timeline.

Requirements

Twitter's news feed displays a user's timeline, including tweets from people they follow, retweets, and mentions. Key considerations for the API design include:

- Data Format: Structuring the response data in a way that's easy to parse.
- Pagination: Managing large volumes of tweets.
- Real-Time Updates: Allowing users to fetch the latest tweets.
- Mentions: Including tweets where the user is mentioned.

API Endpoints

Let's take a look at the Twitter's real [Timeline API](#) to get some inspiration.

Get a User's Timeline

Endpoint: `GET https://api.twitter.com/2/users/:id/timelines/reverse_chronological`

Response Format:

The API returns a JSON object containing:

- `data`: An array of tweet objects.
- `meta`: Information about the response including pagination info.

Sample Response:

```
{ "data": [ { "id": "1524796546306478083", "edit_history_tweet_ids":
["1524796546306478083"], "text": "Today marks the launch of Devs in the Details,
a technical video series made for developers by developers building with the
Twitter API. 🚀nnIn this premiere episode, @jessicagarson walks us through how
she built @FactualCat #WelcomeToOurTechTalknhttps://t.co/nGa8JTQVBJ" }, {
"id": "1524468552404668416", "edit_history_tweet_ids": ["1524468552404668416"],
"text": "📺 Join @jessicagarson @alanbenlee and @i_am_daniele tomorrow, May 12 |
5:30 ET / 2:30pm PT as they discuss the future of bots https://t.co/sQ2bI01fz6"
}, { "id": "1523757705436958720", "edit_history_tweet_ids":
["1523757705436958720"], "text": "Do you make bots with the Twitter API? 🤖
nnJoin @jessicagarson @alanbenlee and @iamdaniele on Thursday, May 12 | 5:30 ET
/ 2:30pm PT as they discuss the future of bots and answer any questions you
might have. 🗣️📅nhttps://t.co/2uVt7hCcdG" }, { "id": "1522642324781633536",
"edit_history_tweet_ids": ["1522642324781633536"], "text": "If you'd like to
apply, or would like to nominate someone else for the program, please feel free
to fill out the following form:nhttps://t.co/LUuWj24HLu" }, { "id":
"1522642323535847424", "edit_history_tweet_ids": ["1522642323535847424"],
"text": "We've gone into more detail on each Insider in our forum post. nnJoin
us in congratulating the new additions! 🎉nhttps://t.co/0r5maYEjPJ" } ],
"meta": { "result_count": 5, "newest_id": "1524796546306478083", "oldest_id":
"1522642323535847424", "next_token":
"7140dibdn0w9c7btw421dyz6jism75z99gyxd8egarsc4", "previous_token":
"7140dibdn0w9c7btw421dyz6jism75z99gyxd8egarsc4" } }
```

A quick note about `edit_history_tweet_ids` in case you are curious. Twitter recently added a feature to edit tweets. And this field stores the unique identifiers indicating all versions of an edited Tweet.

Pagination

Pagination is implemented using the `next_token` and `previous_token` parameters. The client can use these parameters to fetch additional pages of tweets.

Versioning

`api.twitter.com/2` suggests API is at version 2. The version is included in the request URL to allow for future API changes without breaking clients.

Getting Mentions

Endpoint: `GET /2/users/:id/mentions`

"Returns Tweets mentioning a single user specified by the requested user ID. By default, the most recent ten Tweets are returned per request. Using pagination, up to the most recent 800 Tweets can be retrieved."

Sample Response:

```
{ "data": [ { "id": "1375152598945312768", "edit_history_tweet_ids":
["1375152598945312768"], "text": "@LeBraat @TwitterDev @LeGuud There's enough
```

```
@twitterdev love to go around, @LeBraat" }, { "id": "1375152449594523649",
"edit_history_tweet_ids": ["1375152449594523649"], "text": "Apparently I'm not
the only one (of my test accounts) that loves @TwitterDev nn@LeStaaache @LeGuud"
}, { "id": "1375152043455873027", "edit_history_tweet_ids":
["1375152043455873027"], "text": "Can I join this @twitterdev love party?!" }, {
"id": "1375151947360174082", "edit_history_tweet_ids": ["1375151947360174082"],
"text": "I love me some @twitterdev too!" }, { "id": "1375151827189137412",
"edit_history_tweet_ids": ["1375151827189137412"], "text": "This is a test, but
also a good excuse to express my love for @TwitterDev 🥰" } ], "meta": {
"oldest_id": "1375151827189137412", "newest_id": "1375152598945312768",
"result_count": 5, "next_token": "7140dibdnow9c7btw3w3y5b6jqjnk3k4g5zkmfkvqwa42"
} }
```

Pagination and versioning are implemented in the same way as the timeline API.

Hopefully, this example gives you some ideas on how to design your own API. It's a rather basic concept so let's move on and explore more advanced topics.